MIL-P-19131(SHIPS)
AMENDMENT - 3
1 October 1962
SUPERSEDING
Amendment - 2
5 June 1958

MILITARY SPECIFICATION

PUMPS, ROTARY, POWER DRIVEN, MISCELLANEOUS

This amendment forms a part of Military Specification MIL-P-19131 (SHIPS), 23 January 1956.

Page 1, paragraph 1.2: Add:
"Type II - Screws with timing gears.
"Type X - Vane (sliding)."

Pages 1, 2, 3 and 4, paragraph 2.1: Delete reference to Military Specifications "MIL-T-15071", "MIL-N-17506", "NAVY DEPARTMENT General Specifications for Inspection of Material" and "General Specifications for Ships: Section S9-1 Welding and Allied Processes.", and add the following:

SPECIFICATIONS

FEDERAL
"QQ-N-286 - Nickel-Copper-Aluminum Alloy, Wrought (K-Monel).

MILITARY
"MIL-M-15071 - Manuals, Equipment and Systems.

STANDARDS

MILITARY
"MIL-STD-278 - Welding and Allied Processes for Machinery for Ships of the United States Navy."

"PUBLICATIONS

"BUREAU OF SHIPS
"NAVSHIPS 250-637-2 - Instructions for Torch Brazing of Ferrous and Non-Ferrous Piping.
"NAVSHIPS 250-648-8 - Inspection and Test of Silver-Brazed Piping Systems."

Page 5, table I, column 5: Delete "33 - 100" and substitute "30 - 100".

Page 6, paragraph 3.3.1 item (e): Delete and substitute:

"(e) Satisfactory operation when inclined as follows, unless otherwise specified in the contract or order:
(1) Up to 30 degrees from the normal horizontal position in any direction (permanently inclined).

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(2) With the ship rolling up to 60 degrees from the vertical (time of complete roll cycle, 8 seconds).
(3) With the ship pitching 10 degrees up and down from the normal horizontal plane (time of complete pitch cycle, 4 seconds)."

Page 14: Add as paragraph 3.3.16.7:

"3.3.16.7 When specified (see 6.1) pumps shall be provided with mechanical seals."

Page 15, paragraph 3.3.19: Delete and substitute:

"3.3.19 Welding. - Welding shall be in accordance with Standard MIL-STD-278."

"3.3.19.1 Silver brazed joints shall be in accordance with NAVSHIPS 250-637-2. For joints 1/2 inch and above, only preinserted ring fittings shall be used.

Page 16, table II, column 3, line 7: Delete "MIL-N-17506" and substitute "QQ-N-286".

Page 17, paragraph 3.4.3.1: Delete and substitute:

"3.4.3.1 Materials for class N-4 and 0-3 pumps shall be the same as listed in table II except that commercial materials may be used in place of Government specification materials."

Page 17: Add as paragraph 3.4.3.3:

"3.4.3.3 Materials of the vanes of types IX and X pumps shall be that most suitable for classes N-2 and 0-2 services and shall be as proposed by the manufacturer and as approved by the bureau or agency concerned."

Page 18, paragraph 3.5.3.1: Delete formula "$1.5=(\frac{\text{above formula hp.}}{4})$" and substitute:

"$1.5-(\frac{\text{above formula hp.}}{4})$"

Page 21, paragraph 3.8.1: Delete and substitute:

"3.8.1 Manuals shall be in accordance with Specification MIL-M-15071."

Page 21, paragraph 3.8.5, line 5: Delete "MIL-T-15071" and substitute "MIL-M-15071".

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Page 22, table III: Delete and substitute:

"Table III - Onboard repair parts

Item	Name of part	Quantity per set
a b c d e	Ball or roller bearings Oil seals Packing, if special, and packing spacers Coupling bushings, pins, springs, and other wearing parts Mechanical seals	100 percent complete replacement for all installed units per ship
f g	Sleeve, sliding surface bearings Cams, slides, slide pins, housings or liners, vanes (set) and similar wearing parts	One to four units per ship 100 percent replacement for one pump Over four units per ship 100 percent replacement for two pumps
h i	Timing gears, reduction gears and pinions Complete rotating element including bearings and complete coupling	100 percent replacement for one pump

Page 22, paragraph 3.9.2.1, line 1: Delete "item (f)" and substitute "item (g)".

Page 23, paragraph 3.9.2.2, line 1: Delete "item (h)" and substitute "item (i)".

Page 23, paragraph 3.9.2.3, lines 1 and 2: Delete "items (a) and (g)" and substitute "items (a) and (h)", and in line 3, delete "item (h)" and substitute "item (i)".

Page 25, paragraph 4.2.1: Delete and substitute:

"4.2.1 Qualification tests at a laboratory. - Qualification tests on class N-2 and class 0-2 pumps shall be conducted at a laboratory satisfactory to the Bureau of Ships. These tests shall consist of the tests specified in 4.2.8.

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Page 25, paragraph 4.2.2: Delete and substitute:

"4.2.2 In addition to the tests at the laboratory class N-2 and O-2 pumps may be required to pass the following shipboard service test:

(a) Two or more pumps of the same general design and general operating characteristics shall have given satisfactory service onboard Naval ships for a period of at least two years and shall still be in service on such ships at the time qualification approval is granted."

Page 28, paragraph 4.3.3.1: Delete class "O-3" and substitute class "O-2"; line 2, delete "tested on order" and substitute "tested in order".

Page 28, paragraph 4.3.3.2, line 1: Delete class "H-2" and substitute class "N-2"

Page 28, paragraph 4.3.4.1, line 2: Delete "by the representative of the bureau or agency concerned."

Page 30, paragraph 4.3.5, first sentence: Delete and substitute "Sufficient data shall be taken during the tests of 4.3.1, 4.3.2 and 4.3.3 to prepare pump characteristic curves as specified below."

Page 33, paragraph 4.5: Delete and substitute:

"Inspection and test of silver brazed joints in systems designated as hazardous in manual NAVSHIPS 250-648-8 shall be in accordance with the requirements of that manual."

Page 33: Add as paragraph 4.6:

"4.6 Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements."

Page 35, paragraph 6.1 item (h) (9): Delete and substitute:

"(h) (9) Whether mechanical shaft seals are required (see 3.3.16.7)."

Preparing Activity:
Navy - Ships
(Project 4320-N059Sh)

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